Atul Kaple

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Professional Summary

Full Stack Engineer with 2 years of experience building high-throughput web applications and microservices using React, Node.js, Python, MongoDB, and Redis. Specialist in configuring Redis caching layers, structuring high-performance NoSQL data models, and delivering reliable REST APIs.

Technical Skills

Frontend Development: React, Next.js, TypeScript, JavaScript, Tailwind CSS, Material-UI, Redux, Zustand
Backend Development: Node.js, Express.js, Python, REST APIs, Microservices, JWT, Socket.io
Databases & Caching: MongoDB, Mongoose, Redis, MongoDB Atlas
Cloud & DevOps: Docker, AWS (EC2, S3, CloudFront, Elastic Beanstalk), Git, GitHub

Experience

SidLabs LLP **Nov 2024 – Present**
Full Stack Developer *Delhi, India (Remote)*

Multi-Tenant Enterprise Platform (Next.js 15, TanStack Query, Socket.io, Razorpay, ExcelJS)

- Assembled a multi-tenant management portal with Next.js 15, React 19, TypeScript and Node.js/Express, isolating dual-database connections to serve corporate clients with zero cross-tenant contamination.
- Slashed network payload sizes by 72% and achieved sub-10ms cached view paints by deploying offline-first TanStack React Query persistence with client-scoped localStorage caching and HTTP gzip compression.
- Cut redundant database roundtrips by 90% through background sync-status hashing, and dropped dashboard latency from 1,400ms to $<$ 45ms (96% reduction) using in-memory TTL caching and compound indexes.

Patient Community Network (React, TypeScript, Node.js, Socket.io, Redis)

- Developed a privacy-focused social network with polls, media uploads, mood tags, anonymous posting and a ranked feed; Redis caching and cursor pagination support 2,000 concurrent users.
- Indexed posts and comments for semantic search with Transformers.js and MongoDB Atlas Vector Search (RAG), returning summaries that link back to the original posts.
- Gave authors audience controls and applied HIPAA Safe Harbor rules to strip identifying details before posts reach clinical researchers.

Environmental Monitoring Dashboard (Next.js, Leaflet, Node.js, MongoDB)

- Compressed multi-GB environmental datasets into a 4.4MB JSON snapshot through schema encoding and truncation, an 80% size cut that brought dashboard load time under one second.
- Scheduled a node-cron job with Compromise NLP that reads 30+ RSS feeds every 15 minutes, matches 120+ threat keywords, geocodes 200+ Peruvian regions via Nominatim and removes duplicates in MongoDB.
- Drew Leaflet maps with Turf.js 20km buffers and point-in-polygon tests on protected reserves to flag nearby threats and rank the top 5 deforestation hotspots.

Healthcare Research Application (React, TypeScript, Node.js, AWS)

- Created a five-role system (patient, caregiver, clinician, researcher, admin) with an HL7 FHIR R4 assessment module, MongoDB transactions, automatic scoring and S3 video streaming.
- Met HIPAA and GDPR requirements with tiered consent checks, separate MongoDB clusters for health and public data, AWS Cognito MFA and a SHA-256 hash-chained audit log.
- Wrote a de-identification job (AWS KMS decryption, HMAC-SHA256 pseudonyms, small-group suppression with $k < 5$) over 34+ datasets; synced Apple HealthKit data and scheduled medication reminders.

Financial Projection Tool (React, Node.js, ExcelJS, MongoDB)

- Designed a 14-step form flow that produces bank-format Credit Monitoring Arrangement (CMA) projections, loan amortization schedules and 14 annexures for Chartered Accountants.
- Coded an 8,000+ line ExcelJS module that places live formulas across 15+ linked worksheets, with a custom fix for #REF! errors when rows are inserted or removed.
- Issued expiring client links (Node.js crypto tokens, MongoDB TTL index, 72-hour limit) that need no login and refill the multi-step form from saved Zustand state.

Personal Project

Collaborative Whiteboard Application (Next.js, TypeScript, Liveblocks) | Live Link | Demo

- Architected an infinite-canvas collaborative workspace supporting 16+ vector primitives, sticky notes and connector arrows using Next.js 14, React 18, TypeScript and Tailwind CSS.
- Implemented real-time CRDT synchronization via Liveblocks (LiveMap/LiveList) with 16ms throttling (60 FPS), powering conflict-free multi-user editing, ephemeral cursor chat, live reactions, and camera spotlight follow mode.
- Engineered client-side optimizations and secure architecture, utilizing the Ramer-Douglas-Peucker (RDP) algorithm for stroke point reduction, and Clerk + Convex for multi-tenant RBAC with sub-100ms reactive database queries.

Education

Army Institute of Technology **June 2020 – May 2024**
Bachelor of Engineering, Mechanical Engineering | GPA: 8.0 *Pune, Maharashtra*